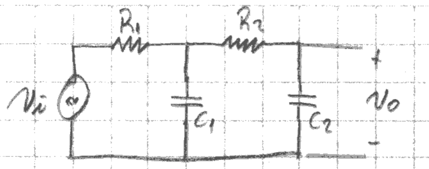
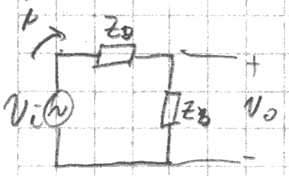
Franco Palombo
401910



$R_1 = 100k$
$R_2 = 182k$
$C_1 = 91pF$
$C_2 = 33pF$

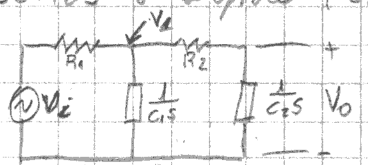
Análisis como cadena de divisores resistivos



$$v_o = i \cdot Z_b$$

$$v_o = \frac{v_i}{Z_a + Z_b} \cdot Z_b$$

Pasamos a Laplace (sin C.I)



$$V_1 = \frac{V_i}{R_1 + \frac{1}{C_1 s}} \cdot \frac{1}{C_1 s} \qquad V_o = \frac{V_1}{R_2 + \frac{1}{C_2 s}} \cdot \frac{1}{C_2 s}$$

$$V_o = V_i \cdot \frac{1}{\left(R_1 + \frac{1}{C_1 s}\right) C_1 s} \cdot \frac{1}{\left(R_2 + \frac{1}{C_2 s}\right) C_2 s}$$

$$\frac{V_o}{V_i} = H_{(s)} = \frac{1}{R_1 C_1 s + 1} \cdot \frac{1}{R_2 C_2 s + 1}$$

$$H_{(s)} = \frac{1}{R_1 C_1 R_2 C_2} \cdot \frac{1}{\left(s + \frac{1}{R_1 C_1}\right)\left(s + \frac{1}{R_2 C_2}\right)}$$

$$H_{(s)} = \frac{18{,}2967 \times 10^9}{(s + 109890{,}1099)(s + 166500{,}1665)}$$

Ceros → NO

Polos → $w = 109890{,}1099 \approx 10^5$
↳ $w = 166500{,}1605 \approx (10{,}5)^5$

$K_{fe} = -10{,}25 \times 10^{-6}\ dB \approx 0dB$

Forma Polar: $s \to jw$

$$H_{(jw)} = \frac{1}{R_1 R_2 C_1 C_2} \cdot \frac{1}{\left(jw + \frac{1}{R_1 C_1}\right)\left(jw + \frac{1}{R_2 C_2}\right)}$$

$$= \frac{1}{R_1 R_2 C_1 C_2} \cdot \frac{1}{-w^2 + \frac{jw}{R_2 C_2} + \frac{jw}{R_1 C_1} + \frac{1}{R_1 C_1 R_2 C_2}}$$

$$= \frac{1}{(1 - R_1 R_2 C_1 C_2 w^2) + jw(R_1 C_1 + R_2 C_2)}$$

Multiplicar por el conj.

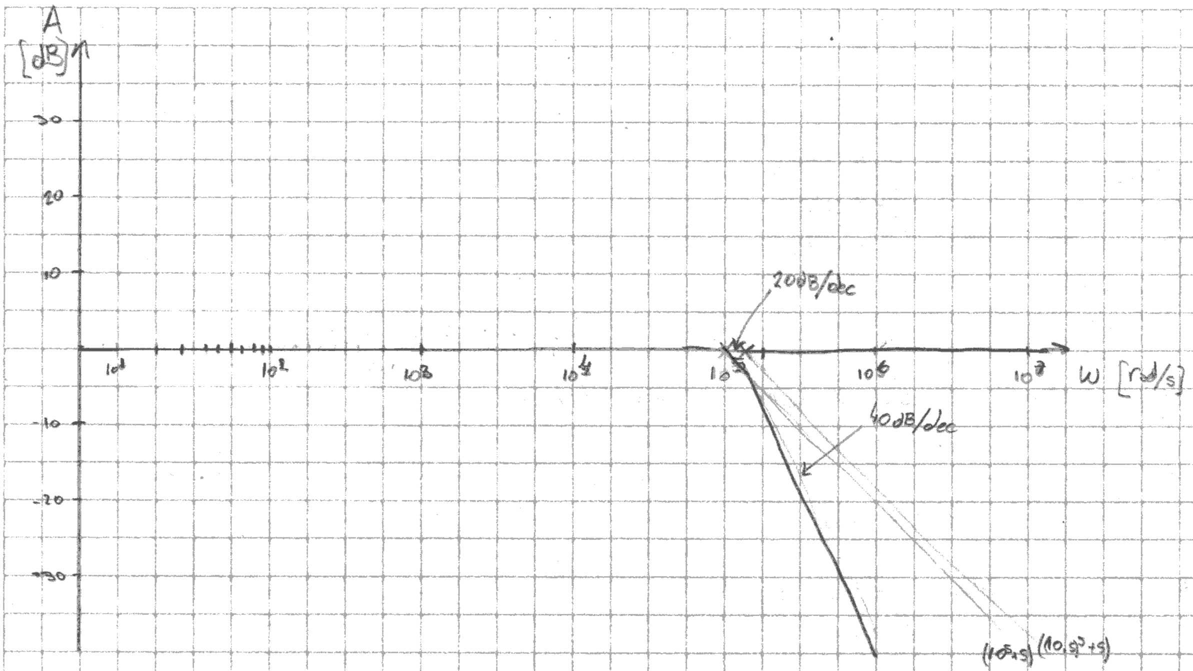
$$= \frac{1 - R_1 R_2 C_1 C_2 w^2 - jw(R_1 C_1 + R_2 C_2)}{(1 - R_1 R_2 C_1 C_2 w^2)^2 + w^2 (R_1 C_1 + R_2 C_2)^2}$$

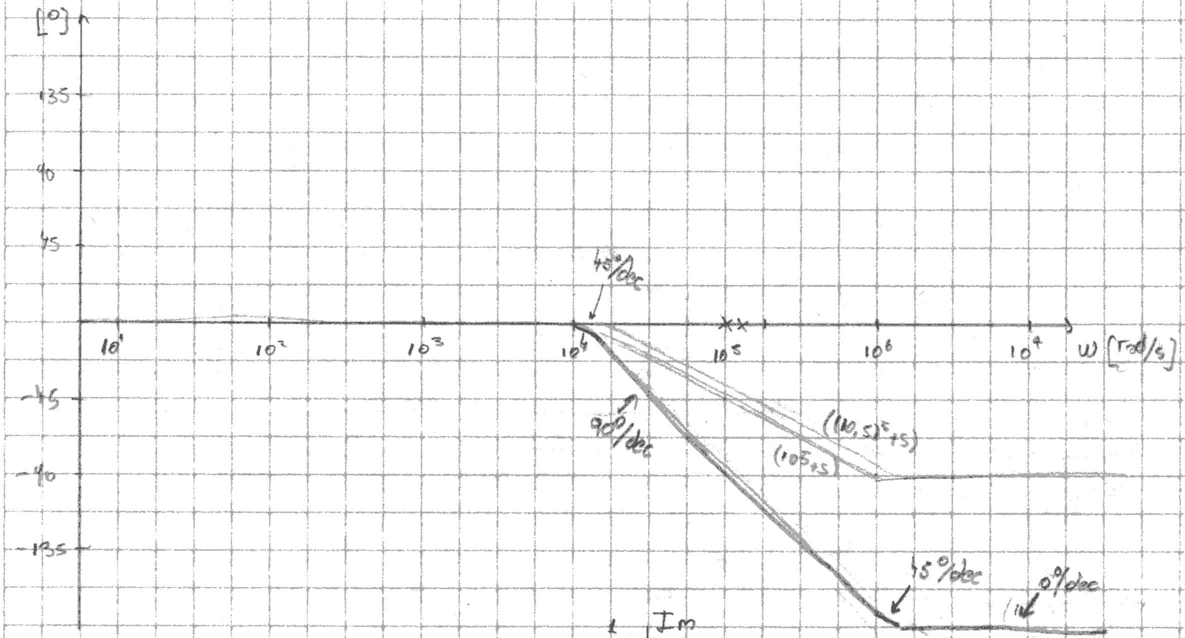
$$Re\{H_{(jw)}\} = \frac{1 - R_1 R_2 C_1 C_2 w^2}{(1 - R_1 R_2 C_1 C_2 w^2)^2 + w^2 (R_1 C_1 + R_2 C_2)^2}$$

$$Re\{H_{(jw)}\} = \frac{1 - w^2 \cdot 54{,}65 \times 10^{-12}}{(1 - w^2 \cdot 54{,}65 \times 10^{-12})^2 + w^2 (9{,}1 \times 10^{-6} + 6 \times 10^{-6})^2}$$

$$Im\{H_{(jw)}\} = \frac{-w(R_1 C_1 + R_2 C_2)}{(1 - R_1 R_2 C_1 C_2 w^2)^2 + w^2 (R_1 C_1 + R_2 C_2)^2}$$

$$Im\{H_{(jw)}\} = \frac{-w(9{,}1 \times 10^{-6} + 6 \times 10^{-6})^2}{(1 - w^2 \cdot 54{,}65 \times 10^{-12})^2 + w^2 (9{,}1 \times 10^{-6} + 6 \times 10^{-6})^2}$$





ω	Re{H}	Im{H}
0	1	0
1	0,99	≈0
10^4	0,98	-0,14
2×10^4	0,93	-0,28
3×10^4	0,85	-0,4
5×10^4	0,65	-0,57
7×10^4	0,44	-0,63
9×10^4	0,25	-0,62
10^5	0,18	-0,6